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# Cardiorespiratory effects of heatwaves: A systematic review and meta-analysis of global epidemiological evidence

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## 16 Abstract

17 **Background:** Heatwaves affect human health and global heatwave-related disease burden will continue to rise as climate change proceeds, but the effects  
18 of heatwaves on cardiovascular and respiratory diseases have not yet been investigated globally and nationally.

19 **Objectives:** This systematic review and meta-analysis aim to quantify heatwave effects on four major health outcomes: cardiovascular and respiratory  
20 morbidity and mortality.

21 **Methods:** We searched PubMed, Scopus, Embase, and Web of Science for relevant studies from database inception to November 2018. Categories of  
22 morbidity included hospital admissions, emergency department visits, and ambulance attendances/call-outs. A random-effects meta-analysis model was  
23 used to pool previous estimates of heatwave effects on mortality and morbidity due to cardiovascular and respiratory diseases. Subgroup analyses by  
24 gender, age, and disease cause were conducted. Sensitivity analyses were performed by the categories of morbidity, heatwave definitions, study design,  
25 and using a leave-one-out cross validation approach. This study is registered with PROSPERO (number: CRD42018101964).

26 **Results:** We identified 54 studies conducted in 20 countries. In total, there were significant associations between heatwaves and cardiovascular mortality  
27 (risk estimates (RE): 1.149, 95% confidence interval (CI): 1.090, 1.210) and respiratory mortality (RE: 1.183, 95%CI: 1.092, 1.282), but the magnitude of these  
28 associations varied across countries and studies. Heatwaves appeared to be marginally associated with cardiovascular and respiratory morbidities (RE:  
29 0.999, 95%CI: 0.996, 1.002,  $p$ -value=0.61 for cardiovascular morbidity; RE: 1.043, 95%CI: 0.995, 1.093;  $p$ -value=0.08 for respiratory morbidity). For mortality,  
30 significant associations were observed for the elderly, ischemic heart disease, stroke, heart failure, and chronic obstructive pulmonary disease. Sensitivity  
31 analyses suggested that these findings were robust.

32 **Conclusion:** Mortality of cardiovascular and respiratory diseases appeared to be more vulnerable to heatwaves in comparison to morbidity. Considering  
33 high heterogeneity detected between studies and limited investigations into subpopulations, more research are required to provide a clearer picture of  
34 how heatwaves affect cardiovascular and respiratory diseases in different settings.

35 **Keywords:** Heatwave; Cardiovascular disease; Respiratory disease; Relative risk

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## 1. Introduction

Heatwave, a sustained period of extremely hot days, has resulted in massive hospitalizations and death toll worldwide, such as 1995 heatwave in USA (Semenza et al., 1996; Semenza et al., 1999), 2003 heatwave in Europe (Robine et al., 2008), 2010 heatwave in Russia (Bagcchi 2015), 2015 heatwave in India (Bagcchi 2015), and 2018 heatwave in Japan (The Lancet 2018). Heatwaves are often regarded as the deadliest natural disaster in Australia and Europe (Coates et al., 2014; Forzieri et al., 2017), which kill more people than all other natural disasters combined. The health impact of heatwaves is expected to rise in the context of population aging and climate change (Guo et al., 2018; Forzieri et al., 2017; Meehl and Tebaldi 2004; Barriopedro et al., 2010).

Although the underlying mechanisms behind health effects of heatwaves have not yet been well understood, many pathophysiological responses associated with extremely high temperatures have been postulated to be responsible for increases in cardiovascular and respiratory diseases (Seltenrich 2015; Gasparrini et al., 2015). For example, exposure to extremely high temperatures could induce elevated heart rate, blood viscosity and plasma cholesterol levels, reduced cerebral perfusion, attenuated vasoconstrictor responsiveness, alternation of fluid and electrolytic balance, and airway hyper-responsiveness (Keatinge et al., 1986; Seltenrich 2015; Lim et al., 2015), which could trigger cardiovascular and respiratory symptoms. Besides, people with pre-existing chronic illnesses such as cardiovascular and respiratory diseases are particularly vulnerable to heatwaves (Seltenrich 2015; Kenny et al., 2010). Hence, cardiovascular and respiratory diseases, the most common medical conditions, would be more likely to be affected by heatwaves.

Although numerous prior studies have examined heatwave effects on cardiovascular and respiratory outcomes, available results are inconsistent (Sherbakov et al., 2018; Phung et al., 2017; Nitschke et al., 2007; Hertel et al., 2009; Nitschke et al., 2011; Sun et al., 2014), provoking uncertainty for researchers and policy makers in understanding and taking precautions against heatwaves. Given the massive burden of disease caused by cardiovascular and respiratory diseases (GBD 2017 Disease and Injury Incidence and Prevalence Collaborators. 2018) and the increasing frequency of heatwaves in the future (Meehl and Tebaldi 2004), it is essential to systematically review available evidence on the associations of heatwaves with cardiovascular and respiratory diseases.

The objective of this systematic review and meta-analysis was to quantify the global and national associations between heatwaves and four major health outcomes (cardiovascular and respiratory morbidity and mortality).

## 2. Methods

This study was conducted and reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analysis guidelines (PRISMA checklist can be found in the supplementary material). The study protocol was registered with PROSPERO (number: CRD42018101964) and available online via <https://www.crd.york.ac.uk/PROSPERO/index.php>.

### 2.1. Search strategy and selection criteria

We searched PubMed, Scopus, Embase, and Web of Science for studies published in English from database inception to November 13, 2018. A number of keywords on heatwaves, cardiovascular disease, and respiratory disease were attempted. Full search terms and search strategy were provided in the supplementary material (p3-p4). We also examined papers identified from the reference lists of included papers to make sure all relevant papers were included.

Studies were included if they met the following criteria. Firstly, the outcome of interest was at least one of four health events, including cardiovascular and respiratory morbidity and mortality. The categories of morbidity included hospital admissions, emergency department visits/presentations, and ambulance attendances/call-outs/transport. Except for ambulance attendances/call-outs/transport, all other morbidity and mortality outcomes must be confirmed according to International Classification of Diseases (ICD), 9th or 10th revision. Secondly, risk estimates (RE) and 95% confidence interval (CI) of heatwaves were reported in terms of relative risk (RR) or odds ratio (OR) or incidence rate ratio (IRR) for morbidity and mortality. Exact figures of RE were shown in studies, or (if not available) could be calculated following the previously used methods (Ma et al., 2011; Rytty et al., 2016; and see the supplementary material, p5).

We also excluded studies in which heatwave definitions were not given or a heatwave was defined as daily temperatures above a threshold without considering the duration (in days) of high temperatures because a heatwave event is typically characterized by high temperatures lasting for several days. Studies that merely investigated the role of characteristics of heatwave such as heatwave effects by duration, rather than heatwave days versus non-heatwave days, were excluded. In order to meta-analyse previous estimates for cause-specific disease, we only considered the following subgroups: ischemic heart disease, stroke, heart failure, chronic obstructive pulmonary disease (COPD) and asthma. As such, we excluded studies that examined only one disease category, which cannot be included in any subgroup.

## 2.2. Data analysis

Two reviewers (JC and ZX) independently screened each study's title and abstract for eligibility. For potentially eligible studies, a full paper was retrieved and screened. Disagreements were resolved through discussion.

For each selected study, we extracted the following information: disease diagnostic criteria, categories of morbidity, authors, study period, study design, study sites, controlled confounders, heatwave definitions (e.g., temperature metrics, intensity, and duration), and effect estimates for overall and added heatwave effects. Main heatwave effects indicate temperature effects from single day's high temperature during heatwaves and added heatwave effects indicate temperature effects from prolonged period of high temperature during heatwaves (Gasparrini and Armstrong 2011; Turner et al., 2013). Overall heatwave effects represent both the main and the added heatwave effects, normally derived from time-series regression models that treat heatwave as binary variable and do not control for temperature (Gasparrini and Armstrong 2011; Cheng et al., 2018).

To obtain the summary estimates of heatwave effects, all measures of RE, including RR, OR, and IRR were pooled using a random-effects model with the empirical Bayes method (Viechtbauer 2010), because the heterogeneity ( $p < 0.05$ ) was observed in most cases (Newell et al., 2017; Cheng et al., 2018).

Our meta-analysis consisted of four stages. Firstly, for each study/country, we generated an RE to measure the overall heatwave effects for each of four outcomes (cardiovascular and respiratory morbidity and mortality). Some studies reported multiple RE such as at different lag days, for different study

periods, and by gender and age group, but did not provide an overall estimate. In this case, all RE in the study were pooled to obtain an overall estimate using above-mentioned meta-analysis method. Some studies reported main and added heatwave effects, and in this case both effects were pooled to obtain the overall heatwave effects. When studies only reported added heatwave effects, they were included in the overall heatwave effect analysis, as well as added heatwave effect analysis. We chose the adjusted effect estimates over the crude effect estimates if they both were reported, and effect estimates from the model with the best goodness-of-fit were adopted if multiple models were used. For multi-country studies, effect estimates were extracted or calculated for each individual country.

Secondly, summary estimates for four different outcomes were obtained by pooling the study-/country-specific RE derived in the first stage. The same analysis was further conducted for each country individually, for extraordinary heatwave events featured by record-long duration or record-high temperatures such as 2003 heatwave in Europe, and for added heatwave effects. The purpose of this stage was to obtain preliminary estimates on country-specific heatwave effects, to check whether very unusual heatwaves had different impacts on cardiovascular or respiratory disease, and to assess whether there was any additional risk from the prolonged period of high temperature.

Thirdly, we conducted analysis for subpopulations by gender, age group (young [0-64 years] and the elderly [65+ years]), disease cause, and categories of morbidity. Some studies focused on populations above 14 or 18 or 50 years of age and this population was considered as the all-age group in the meta-analysis. Similarly, the population group of above 60 or 75 years of age was included in the elderly group (65+) if results for the elderly group (65+) were not reported in studies. Such classification has been used in a previous meta-analysis assessing global association between cold spells and mortality (Ryti et al., 2016). Because the cut-off for different age groups is diverse across previous studies, we here sought to use available data to generally identify vulnerable population, especially for the elderly that constitute the largest proportion of cardiovascular and respiratory deaths.

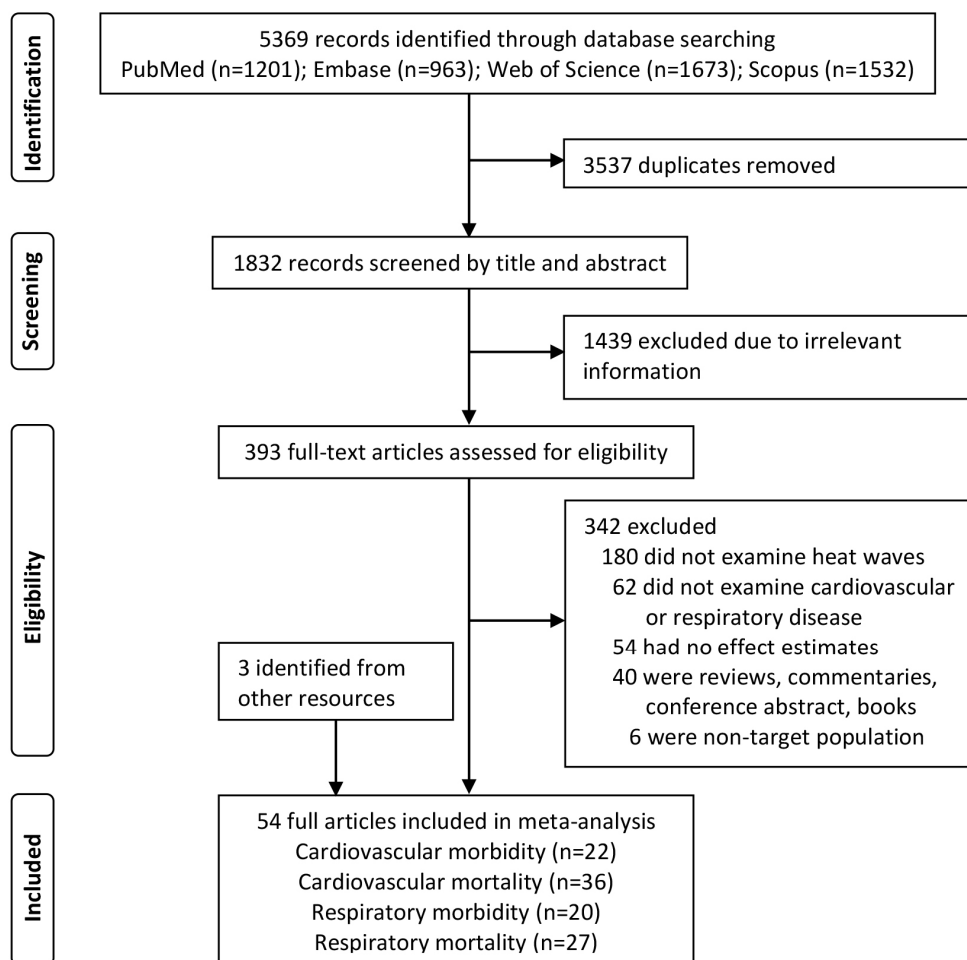
Finally, we did several sensitivity analyses to check the robustness of our findings, including repeated meta-analysis for different categories of morbidity, by heatwave definitions, by study design, splitting the study period before and after the year 2003, using a leave-one-out cross validation approach (leaving out one study/observation at a time), and excluding studies that only examined the added effects of heatwaves (Xu et al., 2016; Barnett et al., 2005).

The statistical significance of the pooled estimates was tested through the Wald test. Heterogeneity between studies was quantified using the  $I^2$  statistic (low 0-25%; moderate 25%-50%; and high >50%) (Sedgwick 2015). Publication bias was assessed by both visually inspecting funnel plots and the Egger test (Egger et al., 1997).

We did all statistical analyses in R software (version 3.4.0) using the package metafor (Viechtbauer 2010; Cheng et al., 2018). In all cases, a two-sided  $p$ -value of 0.05 or less was considered statistically significant.

### 3. Results

Initially, we found 5,369 records, of which 3,537 were duplicates and removed (Fig. 1). We further excluded 1,439 records due to irrelevant title and abstract. 393 full-text articles were screened for eligibility, of which 342 articles were excluded based on inclusion and exclusion criteria. Another three articles were identified after examining the reference lists of included papers. 54 articles across 20 countries were included in the final analysis, including 22 studies on cardiovascular morbidity, 36 on cardiovascular mortality, 20 on respiratory morbidity, and 27 on respiratory mortality (overlaps exist because some studies examined multiple outcomes at the same time). The characteristics of included studies and diagnostic criteria for cardiovascular and respiratory diseases were shown in Tables S1-S5.



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140 **Figure 1:** PRISMA flow diagram of literature search. PRISMA: Preferred Reporting Items for Systematic Reviews and Meta-Analyses

### 3.1. Effects of heatwaves on cardiovascular morbidity

Pooling data from 18 studies demonstrated a statistically non-significant association between heatwaves and cardiovascular morbidity ( $p=0.61$ ) (Fig. 2A). Country-specific summary estimates showed that, except for China, non-significant associations were seen in the five other countries (Australia, South Korea, Spain, USA, and Vietnam). There was low heterogeneity ( $I^2=9.37\%$ ) for the overall heatwave effects group, and low to moderate heterogeneity for four country-specific groups ( $I^2=0\%$  in Australia;  $I^2=37.53\%$  in South Korea;  $I^2=17.65\%$  in USA;  $I^2=35.88\%$  in Vietnam).

Stratified analyses showed that the summary RE was statistically non-significant for all subgroups (Fig. 3), including the males ( $p=0.50$ ) and females ( $p=0.39$ ), the elderly ( $p=0.15$ ) and young group ( $p=0.46$ ), ischemic heart disease ( $p=0.93$ ), stroke ( $p=0.87$ ), and heart failure ( $p=0.16$ ).

There were four extraordinary heatwaves investigated in USA and Australia between 1995 and 2009. Pooled results showed a statistically positive association between extraordinary heatwaves and cardiovascular morbidity (RE: 1.06, 95%CI: 1.01, 1.10;  $p=0.02$ ), with moderate heterogeneity ( $I^2=32.65\%$ ) (Fig. 4).

The pooled results suggested significant added effects of heatwaves on total cardiovascular disease (RE: 1.00, 95%CI: 1.00, 1.01;  $p=0.01$ ) and stroke (RE: 1.02, 95%CI: 1.01, 1.04;  $p<0.01$ ), but non-significant added effects of heatwaves on the remaining subgroups (males and females, and the elderly and young group, ischemic heart disease, and heart failure) (Fig. S1A).

### 3.2. Effects of heatwaves on cardiovascular mortality

There were 27 studies investigating the association between heatwaves and cardiovascular mortality (Fig. 2B). In total, we found significantly positive association (RE: 1.15, 95%CI: 1.09, 1.21;  $p<0.01$ ) and high heterogeneity ( $I^2=99.43\%$ ). This association was also positive in eight countries (Brazil, China, France, Iran, Netherlands, Russia, Thailand, and USA), with RE between 1.05 and 1.40. However, the five other countries (Australia, Germany, Philippines, Slovenia, and Philippines) observed non-significant heatwave effects. High heterogeneity was evident for four country-specific subgroups ( $I^2=92.96\%$  in Australia;  $I^2=89.86\%$  in China;  $I^2=70.39\%$  in South Korea;  $I^2=98.95\%$  in USA).

**(A) Cardiovascular morbidity**

| Author (s) and year                                                | No. of sites | Study period | RR (95% CI)         |
|--------------------------------------------------------------------|--------------|--------------|---------------------|
| <b>Australia</b>                                                   |              |              |                     |
| Nitschke et al. (2007)                                             | 1            | 1993-2006    | 0.982 (0.948-1.018) |
| Mayner et al. (2010)                                               | 1            | 2008-2009    | 1.060 (0.630-1.760) |
| Nitschke et al. (2011)                                             | 1            | 1993-2009    | 1.059 (0.969-1.157) |
| Wang et al. (2012)                                                 | 1            | 1996-2005    | 1.040 (0.870-1.230) |
| Williams et al. (2012)                                             | 1            | 1983-2006    | 1.017 (0.953-1.086) |
| Turner et al. (2013)                                               | 1            | 2000-2007    | 1.009 (0.978-1.042) |
| Toloo et al. (2014)                                                | 1            | 2000-2008    | 0.980 (0.920-1.040) |
| <b>Subtotal: <math>I^2 = 0\%</math>, <math>P = 0.94</math></b>     |              |              | 1.001 (0.981-1.021) |
| <b>China</b>                                                       |              |              |                     |
| Ma et al. (2011)                                                   | 1            | 2005-2008    | 1.080 (1.050-1.110) |
| <b>South Korea</b>                                                 |              |              |                     |
| Son et al. (2014)                                                  | 8            | 2003-2008    | 0.957 (0.856-1.071) |
| Heo et al. (2019)                                                  | 16           | 2011-2014    | 1.078 (0.931-1.248) |
| <b>Subtotal: <math>I^2 = 37.53\%</math>, <math>P = 0.92</math></b> |              |              | 1.006 (0.896-1.129) |
| <b>Spain</b>                                                       |              |              |                     |
| Porrijoan et al. (2017)                                            | 1            | 2006-2013    | 0.961 (0.852-1.083) |
| <b>USA</b>                                                         |              |              |                     |
| Semenza et al. (1999)                                              | 1            | 1994-1995    | 1.100 (0.880-1.320) |
| Knowlton et al. (2009)                                             | 1            | 2006         | 1.015 (1.005-1.025) |
| Sheridan and Lin (2014)                                            | 1            | 1991-2004    | 0.981 (0.959-1.002) |
| Chen et al. (2017)                                                 | 1            | 1945-2012    | 1.000 (1.000-1.001) |
| Sherbakov et al. (2018)                                            | 1            | 1999-2009    | 0.990 (0.985-0.995) |
| <b>Subtotal: <math>I^2 = 17.65\%</math>, <math>P = 0.43</math></b> |              |              | 0.999 (0.996-1.002) |
| <b>Vietnam</b>                                                     |              |              |                     |
| Phung et al. (2016)                                                | 1            | 2004-2013    | 1.020 (0.971-1.072) |
| Phung et al. (2017)                                                | 25           | 2002-2015    | 0.987 (0.972-1.001) |
| <b>Subtotal: <math>I^2 = 35.88\%</math>, <math>P = 0.69</math></b> |              |              | 0.995 (0.968-1.022) |
| <b>Overall: <math>I^2 = 9.37\%</math>, <math>P = 0.61</math></b>   |              |              | 0.999 (0.996-1.002) |

**(B) Cardiovascular mortality**

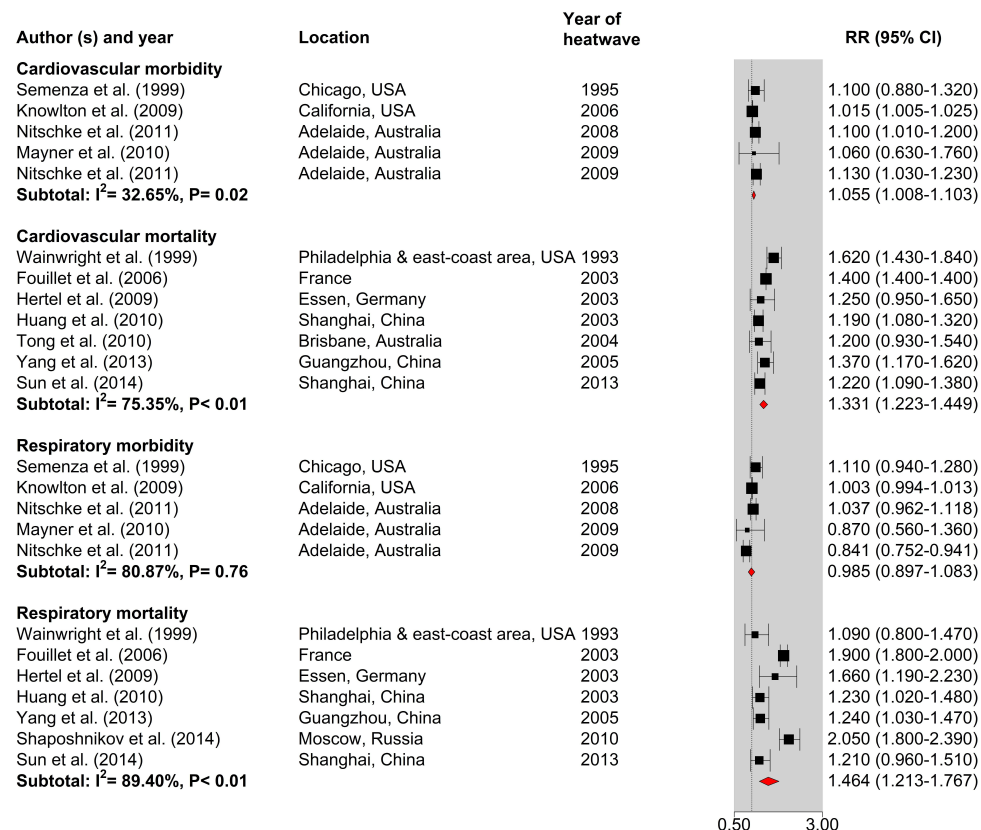
| Author (s) and year                                                   | No. of sites | Study period | RR (95% CI)         |
|-----------------------------------------------------------------------|--------------|--------------|---------------------|
| <b>Australia</b>                                                      |              |              |                     |
| Tong et al. (2010)                                                    | 1            | 2001-2004    | 1.200 (0.930-1.540) |
| Nitschke et al. (2007)                                                | 1            | 1993-2006    | 0.900 (0.840-0.960) |
| Wang et al. (2012)                                                    | 1            | 1996-2004    | 1.890 (1.440-2.480) |
| <b>Subtotal: <math>I^2 = 92.96\%</math>, <math>P = 0.30</math></b>    |              |              | 1.249 (0.817-1.908) |
| <b>Brazil</b>                                                         |              |              |                     |
| Son et al. (2016)                                                     | 1            | 1996-2000    | 1.076 (1.025-1.130) |
| <b>China</b>                                                          |              |              |                     |
| Huang et al. (2010)                                                   | 1            | 2003         | 1.190 (1.080-1.320) |
| Yang et al. (2013)                                                    | 1            | 2003-2006    | 1.290 (1.073-1.552) |
| Sun et al. (2014)                                                     | 1            | 2008-2013    | 1.220 (1.090-1.380) |
| Zeng et al. (2014)                                                    | 1            | 2006-2010    | 1.193 (1.095-1.301) |
| Chen et al. (2015a)                                                   | 1            | 2007-2013    | 1.349 (1.216-1.498) |
| Ma et al. (2015)                                                      | 66           | 2006-2011    | 1.077 (1.047-1.107) |
| Dong et al. (2016)                                                    | 1            | 2006-2009    | 1.048 (1.015-1.083) |
| Han et al. (2017)                                                     | 1            | 2011-2014    | 1.030 (1.000-1.060) |
| Yin et al. (2018)                                                     | 69           | 2013-2015    | 1.180 (1.110-1.240) |
| Yang et al. (2019)                                                    | 31           | 2007-2013    | 1.130 (1.080-1.190) |
| <b>Subtotal: <math>I^2 = 89.86\%</math>, <math>P &lt; 0.01</math></b> |              |              | 1.148 (1.089-1.210) |
| <b>France</b>                                                         |              |              |                     |
| Fouillet et al. (2006)                                                | na           | 2000-2003    | 1.400 (1.400-1.400) |
| <b>Germany</b>                                                        |              |              |                     |
| Hertel et al. (2009)                                                  | 1            | 2000-2006    | 1.250 (0.950-1.650) |
| <b>Iran</b>                                                           |              |              |                     |
| Ahmadnezhad et al. (2013)                                             | 1            | 2001-2011    | 1.046 (1.039-1.054) |
| <b>Netherlands</b>                                                    |              |              |                     |
| Huyten et al. (2001)                                                  | 1            | 1979-1997    | 1.088 (1.028-1.147) |
| <b>Philippines</b>                                                    |              |              |                     |
| Seposo et al. (2017)                                                  | 4            | 2006-2010    | 0.999 (0.967-1.033) |
| <b>Russia</b>                                                         |              |              |                     |
| Revich and Shaposhnikov (2008)                                        | 1            | 2000-2006    | 1.150 (1.050-1.260) |
| <b>Slovenia</b>                                                       |              |              |                     |
| Percic et al. (2018)                                                  | na           | 2003 & 2015  | 1.026 (0.823-1.279) |
| <b>South Korea</b>                                                    |              |              |                     |
| Lee et al. (2016)                                                     | 1            | 1992-2012    | 1.122 (1.020-1.233) |
| Heo et al. (2019)                                                     | 16           | 2011-2014    | 1.025 (1.007-1.043) |
| <b>Subtotal: <math>I^2 = 70.39\%</math>, <math>P = 0.19</math></b>    |              |              | 1.059 (0.973-1.153) |
| <b>Thailand</b>                                                       |              |              |                     |
| Huang et al. (2018)                                                   | 60           | 1999-2008    | 1.087 (1.069-1.106) |
| <b>USA</b>                                                            |              |              |                     |
| Wainwright et al. (1999)                                              | 5            | 1993         | 1.620 (1.430-1.840) |
| Anderson and Bell (2009)                                              | 107          | 1987-2000    | 1.088 (1.055-1.122) |
| Sheridan and Lin (2014)                                               | 1            | 1991-2004    | 1.191 (1.157-1.227) |
| <b>Subtotal: <math>I^2 = 98.95\%</math>, <math>P = 0.04</math></b>    |              |              | 1.272 (1.008-1.605) |
| <b>Overall: <math>I^2 = 99.43\%</math>, <math>P &lt; 0.01</math></b>  |              |              | 1.149 (1.090-1.210) |

**Figure 2:** Summary of the overall association of heatwaves with cardiovascular morbidity (A) and cardiovascular mortality (B). Pooled risk estimates were calculated using random effects meta-analyses. P indicates the significant test for the pooled estimates.

163

164 **Figure 3:** The overall association of heatwaves with cardiovascular morbidity by gender, age, and disease cause. Pooled risk estimates were calculated using  
165 random effects meta-analyses. P indicates the significant test for the pooled estimates.

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**Figure 4:** Summary of the association of extraordinary heatwaves with four outcomes: cardiovascular morbidity, cardiovascular mortality, respiratory morbidity, and respiratory mortality. Pooled risk estimates were calculated using random effects meta-analyses. Extraordinary heatwaves indicate the heatwave with unusually long duration or record high temperature such as 2003 heatwave in Europe. P indicates the significant test for the pooled estimates.

172 Subgroup analysis by gender and age revealed positive associations for the males (RE:1.05, 95%CI: 1.02, 1.09;  $p < 0.01$ ), and the elderly (RE: 1.20, 95%CI:  
173 1.08, 1.32;  $p < 0.01$ ) (Fig. 5). For different disease causes, the association was positive for ischemic heart disease (RE:1.23, 95%CI: 1.07, 1.42;  $p < 0.01$ ), stroke  
174 (RE:1.19, 95%CI: 1.04, 1.36;  $p = 0.01$ ), and heart failure (RE: 1.10, 95%CI: 1.04, 1.18;  $p < 0.01$ ). Low and high heterogeneity were detected across different  
175 subgroups.

176 Five extraordinary heatwaves were studied across five countries (Australia, China, France, German, and USA) during the period 1993-2013 (Fig. 4). The  
177 pooled RE was 1.33 (95%CI: 1.22, 1.45;  $p < 0.01$ ), with high heterogeneity ( $I^2 = 75.35\%$ ).

178 Similar to overall association between heatwaves and cardiovascular mortality, added heatwave effects were significant for the total cardiovascular  
179 mortality (RE:1.09, 95%CI: 1.04, 1.14;  $p < 0.01$ ), the males (RE:1.05, 95%CI: 1.02, 1.09), the elderly (RE:1.06, 95%CI: 1.01, 1.11), ischemic heart disease  
180 (RE:1.15, 95%CI: 1.05, 1.25;  $p < 0.01$ ), and stroke (RE:1.14, 95%CI: 1.03, 1.26;  $p = 0.01$ ) (Fig. S1B). For females and the young group, heatwave effects were  
181 not significant.

182

183 **Figure 5:** The overall association of heatwaves with cardiovascular mortality by gender, age, and disease cause. Pooled risk estimates were calculated using  
184 random effects meta-analyses. P indicates the significant test for the pooled estimates.

### 3.3. Effects of heatwaves on respiratory morbidity

Pooling data from 17 studies showed a statistically non-significant association between heatwaves and respiratory morbidity ( $p=0.08$ ), with high heterogeneity ( $I^2=98.61\%$ ) observed (Fig. 6A). We observed a significant association in South Korea (RE:1.07, 95%CI: 1.05, 1.10;  $p<0.01$ ), although the non-significant association was evident in four other countries (Australia, China, USA, and Vietnam).

Our gender-, age-, and disease-specific analyses found non-significant associations (neither negative nor protective) in all subgroups (Fig. 7). High heterogeneity ( $I^2>70\%$ ) was observed in the elderly and COPD groups, and low to moderate heterogeneity ( $I^2<50\%$ ) was observed in the young group and asthma group.

Four extraordinary heatwaves that occurred in USA and Australia between 1995 and 2009 were analysed (Fig. 4). A non-significant association between extraordinary heatwaves and respiratory morbidity was found ( $p=0.76$ ).

Similarly, heatwaves had no added effects on the total respiratory morbidity, the young group, COPD, and asthma (Fig. S1C). However, we observed significant added heatwave effects on the elderly (RE: 1.02, 95%CI: 1.00, 1.03;  $p=0.01$ ).

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202 **Figure 6:** Summary of the overall association of heatwaves with respiratory morbidity (A) and respiratory mortality (B). Pooled risk estimates were  
203 calculated using random effects meta-analyses. P indicates the significant test for the pooled estimates.

(A)

| Author (s) and year                                                | No. of sites | Study period | RR (95% CI)         |
|--------------------------------------------------------------------|--------------|--------------|---------------------|
| <b>Male</b>                                                        |              |              |                     |
| Phung et al. (2017)                                                | 25           | 2002-2015    | 0.997 (0.968-1.027) |
| <b>Female</b>                                                      |              |              |                     |
| Phung et al. (2017)                                                | 25           | 2002-2015    | 0.983 (0.949-1.018) |
| <b>Young (0-64)</b>                                                |              |              |                     |
| Nitschke et al. (2007)                                             | 1            | 1993-2006    | 0.991 (0.906-1.085) |
| Mayner et al. (2010)                                               | 1            | 2008-2009    | 0.890 (0.500-1.600) |
| Nitschke et al. (2011)                                             | 1            | 1993-2009    | 0.993 (0.900-1.095) |
| Wang et al. (2012)                                                 | 1            | 1996-2005    | 1.170 (0.820-1.670) |
| Turner et al. (2013)                                               | 1            | 2000-2007    | 1.135 (0.996-1.295) |
| Phung et al. (2017)                                                | 25           | 2002-2015    | 0.962 (0.933-0.991) |
| <b>Subtotal: <math>I^2 = 28.04\%</math>, <math>P = 0.81</math></b> |              |              | 0.994 (0.947-1.043) |
| <b>Elderly (65+)</b>                                               |              |              |                     |
| Nitschke et al. (2007)                                             | 1            | 1993-2006    | 0.938 (0.897-0.981) |
| Mayner et al. (2010)                                               | 1            | 2008-2009    | 0.840 (0.420-1.690) |
| Nitschke et al. (2011)                                             | 1            | 1993-2009    | 0.905 (0.850-0.964) |
| Wang et al. (2012)                                                 | 1            | 1996-2005    | 1.219 (0.899-1.652) |
| Turner et al. (2013)                                               | 1            | 2000-2007    | 1.645 (1.292-2.096) |
| Mastrangelo et al. (2007)                                          | 1            | 2002-2003    | 1.322 (1.160-1.508) |
| Bobb et al. (2014)                                                 | 1943         | 1999-2000    | 1.008 (0.992-1.023) |
| Gronlund et al. (2014)                                             | 200          | 1992-2006    | 1.014 (1.002-1.026) |
| Phung et al. (2017)                                                | 25           | 2002-2015    | 0.988 (0.966-1.010) |
| <b>Subtotal: <math>I^2 = 99.24\%</math>, <math>P = 0.28</math></b> |              |              | 1.072 (0.945-1.216) |

0.50 1.50

(B)

| Author (s) and year                                                | No. of sites | Study period | RR (95% CI)         |
|--------------------------------------------------------------------|--------------|--------------|---------------------|
| <b>COPD</b>                                                        |              |              |                     |
| Semenza et al. (1999)                                              | 1            | 1994-1995    | 1.130 (0.980-1.270) |
| Chen et al. (2017)                                                 | 1            | 1945-2012    | 1.000 (0.999-1.000) |
| <b>Subtotal: <math>I^2 = 70.74\%</math>, <math>P = 0.46</math></b> |              |              | 1.044 (0.932-1.171) |
| <b>Asthma</b>                                                      |              |              |                     |
| Mayner et al. (2010)                                               | 1            | 2008-2009    | 1.000 (0.440-2.300) |
| Son et al. (2014)                                                  | 8            | 2003-2008    | 1.091 (0.813-1.463) |
| Chen et al. (2017)                                                 | 1            | 1945-2012    | 0.998 (0.995-1.001) |
| <b>Subtotal: <math>I^2 = 0\%</math>, <math>P = 0.17</math></b>     |              |              | 0.998 (0.995-1.001) |

0.50 1.50

**Figure 7:** The overall association of heatwaves with respiratory morbidity by gender, age, and disease cause. Pooled risk estimates were calculated using random effects meta-analyses. P indicates the significant test for the pooled estimates.

### 208 **3.4. Effects of heatwaves on respiratory mortality**

209 The association between heatwaves and respiratory mortality were examined in 25 studies from 13 countries (Fig. 6B). We found an overall significant  
 210 association (RE: 1.18, 95%CI: 1.09, 1.28;  $p < 0.01$ ) and high heterogeneity ( $I^2=97.20\%$ ) across studies. Analysis by country suggested significant association in  
 211 seven countries (Brail, China, France, Germany, Netherlands, Russia, and Thailand), and the pooled RE was between 1.10 and 2.05. Non-significant  
 212 association was observed in the remaining six countries (Australia, Iran, Philippines, Slovenia, South Korea, and USA). Evidence of no heterogeneity was  
 213 noted in USA ( $I^2=0\%$ ), but moderate and high heterogeneity was found in Australia ( $I^2=48.62\%$ ), South Korea ( $I^2=45.64\%$ ) and China ( $I^2=68.86\%$ ), respectively.

214 Subgroup analyses showed significant associations between heatwaves and respiratory mortality in the elderly (RE:1.33, 95%CI: 1.16, 1.53;  $p < 0.01$ ), and  
 215 COPD (RE:1.14, 95%CI: 1.08, 1.21;  $p < 0.01$ ), but a non-significant association in the young group ( $p = 0.24$ ) (Fig. 8).

216 After pooling estimates for five extraordinary heatwaves that happened in five countries (China, France, Germany, Russia, USA) between 1993-2013, we  
 217 found a significant association between extraordinary heatwaves and respiratory mortality (RE:1.46, 95%CI: 1.21, 1.77;  $p < 0.01$ ), with high heterogeneity  
 218 ( $I^2=89.40\%$ ) (Fig. 4).

219 For added heatwave effects (Fig. S1D), we found significant associations for total respiratory mortality (RE:1.08, 95%CI: 1.01, 1.16;  $p = 0.03$ ) and COPD  
 220 (RE:1.15, 95%CI: 1.03, 1.29;  $p = 0.02$ ).

### 221 **3.5. Publication bias and sensitivity analysis**

222 After inspecting the funnel plots and performing the Egger tests, we did not find publication bias for analyses of three outcomes (cardiovascular  
 223 morbidity, respiratory morbidity and mortality) (Fig. S2). However, significant publication bias existed for analysis of cardiovascular mortality ( $p < 0.01$ ). We  
 224 performed a series of sensitivity analyses and found the main findings remained stable (Figs. S3-S4 and Tables S6-S8).

225

226 **Figure 8:** The overall association of heatwaves with respiratory mortality by gender, age, and disease cause. Pooled risk estimates were calculated using  
227 random effects meta-analyses. P indicates the significant test for the pooled estimates.

228

229

#### 4. Discussion

Heatwaves are the most lethal weather-related hazard (Forzieri et al., 2017), being a global public health concern in the 21st century considering warming climate and population ageing (Watts et al., 2018). Most heat-related health events are likely to be from cardiovascular or respiratory causes (Hajat et al., 2010; Guo et al., 2018). However, current findings on the associations between heatwaves and cardiovascular and respiratory diseases are mixed in the literature. In order to obtain a clear understanding of the effects of heatwaves on different cardiovascular and respiratory outcomes and identify potential vulnerable population, we systematically reviewed previous relevant studies and meta-analysed reported effect estimates. Our findings suggest that heatwaves in general were associated with increased risk of cardiovascular and respiratory mortality but had no significant effects on cardiovascular and respiratory morbidity. For mortality, significant associations were found for the elderly, ischemic heart disease, stroke, heart failure, and COPD. Noticeably, except for respiratory morbidity, risk of three other outcomes (cardiovascular morbidity and mortality, and respiratory mortality) increased significantly during extraordinary heatwave periods.

To our knowledge, this is the first review to examine heatwave effects on the four most common health outcomes: cardiovascular and respiratory morbidity and mortality. Many reviews and multi-country studies also assessed health impacts of heatwaves (Åström et al., 2011; Kravchenko et al., 2013; Li et al., 2015; Phung et al., 2016; Xu et al., 2016; Guo et al., 2017; Sun et al., 2018; Guo et al., 2018; Campbell et al., 2018), but some of them synthesized previous findings only in a narrative way and many studies on this topic have been published thereafter. In addition, the outcomes examined in previous literature were limited to all-cause/non-accidental mortality, or one particular category of morbidity (e.g., cardiovascular hospitalizations).

Cardiovascular and respiratory health during heatwaves could be monitored by a number of ways, such as hospital admissions, emergency department visits/presentations and ambulance attendances/call-outs/transport. Heatwave effects could differ by categories of morbidity that mirrors different severities of the same outcome. However, this hypothesis was not supported by our meta-analysis, which showed non-significant associations for both cardiovascular and respiratory diseases within any category (Fig. S4). This finding is similar to a previous report that heat (temperature rise) seems not to affect cardiovascular and respiratory morbidity (Turner et al., 2012).

250 In contrast to findings on morbidity (non-significant overall heatwave effects), we found that heatwaves significantly affected both cardiovascular and  
251 respiratory mortality. Adverse heatwave effects on cardiovascular and respiratory mortality were also observed in our subgroup analyses and validated by  
252 our sensitivity analysis (Figs. 5 and 8, Tables S6-S8). The contrasting heatwave effects on mortality and morbidity (from cardiovascular and respiratory  
253 diseases) might be related to a number of factors. Since the heat effects are acute, which could occur within several hours (Bhaskaran et al., 2012; Guo  
254 2017), vulnerable people during heatwaves may die rapidly before they receive timely medical treatment or are admitted to hospital (Turner et al., 2012;  
255 Michelozzi et al., 2009). Hence, analysing morbidity data at a relatively coarse temporal resolution (e.g., daily data) seems to be inappropriate to capture  
256 the acute and transient effects of heatwaves. Another reason is probably due to the divergent effects of heatwave compositions, including main heatwave  
257 effects from single day's high temperature and added heatwave effects from sustained period of high temperature. Since the added heatwave effects are  
258 supposed to be negligible in comparison to the overall heatwave effects (Gasparrini and Armstrong 2011), people may or may not be affected by the  
259 prolonged period of high temperature. There is good evidence that people within a given region or a particular subgroup were found to be sensitive to only  
260 one or both heatwave compositions (Sherbakov et al., 2018; Song et al., 2018; Turner et al., 2013). Therefore, previous analyses of heatwave effects that  
261 failed to consider different heatwave compositions likely obscure the true effects of heatwaves, as shown in Fig. S1 that heatwaves had significant  
262 additional effects on cardiovascular morbidity (all-cause and stroke) and respiratory morbidity (elderly group), while overall heatwave effects on morbidity  
263 from cardiovascular or respiratory disease was not significant (Figs. 2 and 6). The non-significant heatwave effects on cardiovascular and respiratory  
264 morbidity may also be caused by the harvesting effects of heatwaves, a widely reported phenomenon for mortality known as sudden increases in deaths  
265 during the first few days followed by significant decreases in deaths several days later (Cheng et al., 2018). Furthermore, the current investigations into  
266 morbidity effects of heatwaves were carried out in few countries and many other regions such as low-income countries that may be sensitive to heatwaves  
267 have not yet been included, which may prevent us from detecting the adverse effects of heatwaves. Because the pooled estimates for cardiorespiratory  
268 morbidity were marginally significant (i.e., lower limit of confidence interval was very close to the value "1") (Figs. 2 and 6), more investigations are needed  
269 to verify if heatwaves have adverse effects on cardiorespiratory morbidity.

270 As the climate continues to warm, extraordinary heatwaves can be considered as one of the biggest global health threats of the 21st century (Watts et al.,  
271 2018). Our study found that very unusual heatwaves have adverse effects on three outcomes (cardiovascular morbidity and mortality, and respiratory  
272 mortality). Conversely, the occurrence of respiratory morbidity events seems to be little affected by unusual heatwaves, but the available data is only from  
273 two high-income countries (USA and Australia). More of such studies from other regions are clearly needed.

274 We also found the associations of heatwaves with four outcomes varied between countries, with both significant and non-significant associations  
275 observed (Figs. 2 and 6). For example, heatwaves were not associated with increased risk for four outcomes in Australia and for three outcomes in USA  
276 (except for cardiovascular mortality) and South Korea (except for respiratory morbidity). This finding probably suggests that population in some high-  
277 income countries may to some extent have adapted to the local heatwave weather. But our meta-regression analyses did not show significant difference in  
278 risks of cardiovascular and respiratory mortality and cardiovascular morbidity between high-income and middle-income countries (Fig. S5). Moreover,  
279 people in all regions of a country would not be exempt from heatwave impacts, because many factors such as climate, population demographic,  
280 socioeconomic, and infrastructural characteristics could be distinct across a country that cause the heatwave effects spatially variable (Xu et al., 2016;  
281 Phung et al., 2017; Ma et al., 2015).

282 Our meta-analysis by age suggested that heatwaves more likely affect cardiovascular and respiratory mortality among the elderly, which is consistent  
283 with previous findings on all-cause/non-accidental mortality (Zeng et al., 2014; Anderson and Bell 2009; Shaposhnikov et al., 2014). We also observed  
284 increased mortality risk for ischemic heart disease, stroke, heart failure, and COPD. These findings are very useful for health authorities and clinicians when  
285 giving advice to protect the public and high-risk populations from adverse heatwave effects.

286 Several limitations of this study must be acknowledged. Although our report included populations living in a number of countries with a wide range of  
287 climates, cautions must be taken when interpreting the summary estimates because of high heterogeneity observed in many cases, and lack of studies for  
288 some countries and subgroups. While null or low heterogeneity was found in subgroup analyses, high heterogeneity for the heatwave effects on total  
289 cardiovascular and respiratory diseases still existed. The potential reasons are multiple, including distinct climates of study sites, study designs and

statistical methods (Xu et al., 2016). Meta-regression seems to be a sound approach to exploring the resource of heterogeneous effects between studies, but it is hard to accomplish this goal here since there were very few studies with similar characteristics and multi- and single-site studies added uncertainty of extracting necessary information for meta-regression analysis.

Although we performed stratified analyses by heatwave definitions (Fig. S3), it remains unclear if heatwave effects on cardiovascular and respiratory diseases can be modified by its characteristics such as the intensity (the height of temperature), because previous studies did not provide adequate effect estimates. A recent Australian study found evidence of higher mortality risk associated with more intense heatwaves (Cheng et al., 2018). However, a previous meta-analysis did not observe an appreciable trend for the association between heatwave intensity and mortality risk (Xu et al., 2016). More studies are thus needed to investigate how heatwave intensity determines the risk of cardiovascular and respiratory diseases, regionally and nationally.

We were unable to include countries in Africa or the Middle East due to the issue of data availability. It has been shown that people Africa were also sensitive to high temperatures (Kula et al., 2013), causing excessive premature deaths and hospitalizations (Amegah et al., 2016; Scovronick et al., 2018). However, whether heatwaves in Africa and the Middle East affect cardiovascular and respiratory outcomes has been unknown to date. Such studies are thus urgently needed, so that to help gain a deeper understanding of heatwave effects and to better cope with climate change (Kula et al., 2013).

Significant publication bias was observed in the meta-analyses of one (cardiovascular mortality) of four outcomes, but the underlying reasons are unclear. This might result from our language limitation applied to literature search. We also carried out trim and fill method (a technology to identify and correct for funnel plot asymmetry arising from publication bias) to assess if and how many studies are missing from our meta-analysis (Duval and Tweedie 2000). The results showed no difference with the original pooled RE and no evidence of missing study ( $p=0.25$ ).

So far, various heatwave definitions have been used in the health risk assessment, as shown in the present (Table S9) and previous studies (Xu et al., 2016; Guo et al., 2017). Studies also pointed out appreciable variations in the effect estimates across different heatwave definitions (Xu et al., 2016; Chen et al., 2015a; Guo et al., 2017), underlying the great importance of having the reasonable heatwave definition (s). Although it is not feasible to reach a universal or

global definition, defining the nature of a 'heatwave' regionally or nationally is supposed to be achievable (Tong et al., 2015; Xu et al., 2016). There are few efforts made on this topic. In Australia, Tong and colleagues (2015) used the health risk-based metrics for defining a heatwave. They tested a number of heatwave definitions (daily mean temperature exceeded 75-99th percentiles and lasted for at least 2 days), and found that mortality risk started to increase at 95th percentile, sharply increased at 97th percentile, and alarmingly increased at 99th percentile. This heatwave definition method can be widely adopted from morbidity to mortality, and from one city to multiple cities in Australia (Tong et al., 2014; Tong et al., 2015; Cheng et al., 2018). In Spain, researchers defined a heatwave from the statistical perspective, and found the trigger threshold varied across regions (Díaz et al., 2015). The ideal heatwave definition could help early warning and prevention systems to cover the general public, vulnerable populations such as the elderly and those with pre-existing medical conditions, and identify "hot spots" at the lowest cost. This cannot be easily addressed with a single heatwave definition, but requires a comprehensive risk assessment framework such as the tiered heatwave definitions for the whole population and for different subgroups within a given region (Tong et al., 2015).

## 5. Conclusions

This systematic review and meta-analysis demonstrate that, among four studied health outcomes, neither cardiovascular morbidity nor respiratory morbidity was significantly affected by heatwaves in general, whereas heatwaves had apparent effects on both cardiovascular and respiratory mortality. The associations (significant or not) of heatwaves with four outcomes varied with country, gender, and disease cause. For cardiovascular and respiratory mortality, heatwaves were more likely to affect the elderly, including deaths from ischemic heart disease, stroke, heart failure, and COPD. More of such research is still needed, especially for unexplored or less studied regions and subpopulations, in view of global warming and population aging worldwide.

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333 **Conflict of interest**

334 The authors declare no conflict of interest.

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